A larger budget, a board that forgets faster or slower

Simulation 2 (asynchronous activation), setup `task003-symmetric`: three variants against the gold standard

LLM-free simulator; report of 19 October 2026; runs of 15, 16 and 18 October 2026

Same sections, figures and equations as the gold-standard report of 16 October, with the four cases side by side.

How the simulator works. The world, `task_003`, has three candidate allocations; A0 is the correct one. Each of 24 agents is given one true fact and reasons as an exact Bayesian: its *belief* is the exact probability of A0, A1 and A2 given the facts it currently remembers. There are no rounds. Each agent acts at random times, on average once per time unit: it reads q posts drawn at random from the W most recent posts ($W = 48$ in the gold standard) on a shared *board* (never its own), adds their facts to its memory, recomputes its belief, *votes* by probability matching (a vote drawn from its belief) and posts one remembered fact that supports its vote. Every remembered fact is forgotten after a random lifetime (on average a fraction ρ survives one time unit; reading it again renews it). A *controller* with a target, A0 (truth) or A2 (false), acts at random times at rate λ_c until $t = 30$: it reads q_c recent agent posts and, unless the *vote gate* (at least 75% of them from agents voting for its target) or the *proof stop* (its target already proved) silences it, posts the one fact from its pool that best supports its target, at most B messages per episode ($B = 30$ in the gold standard). Agents continue alone until $t = 40$. Every episode is run three times with the same random draws: without a controller (*silent*), with the controller targeting A0 and with the one targeting A2.

The facts at the start. All facts are true. The agents hold, together, a proof of A0; no single agent does, and neither pool can prove its target (Table 1).

Table 1: Facts available at $t = 0$ and what each set implies, $P(\text{A0}, \text{A1}, \text{A2} \mid \text{facts})$.

set of facts	facts	$P(\text{A0}, \text{A1}, \text{A2})$
agents' facts, all together (24 agents, one fact each)	18 distinct	(1, 0, 0): proves A0
A0 pool (controller with target A0)	12, of which 10 new to the agents	(0.963, 0.037, 0)
A2 pool (controller with target A2)	12, of which 10 new to the agents	(0.114, 0, 0.886)
agents' facts + either pool	28 distinct	(1, 0, 0)

The three variants. Each changes one rule of the gold standard and keeps everything else (Table 2).

- **Budget.** In the gold standard the controller may post at most $B = 30$ messages before $t = 30$; from $\lambda_c = 2$ upward it uses almost all of them, and at $\lambda_c = 8$ it runs out near $t = 4$. With $B = 100$ it can keep posting until $t = 30$ for $\lambda_c \lesssim 3$ and runs out near $t = 12.5$ at $\lambda_c = 8$; the vote gate can still silence it. This separates the rate from the total effort.
- **Board window.** Nothing is ever deleted from the board, but every reader (agent or controller) draws only from the W newest posts it may read. The 24 agents post about 24 times per time unit, so in the gold standard ($W = 48$) a post stays readable for about 2 time units unless someone posts its fact again. With $W = 24$ it stays about 1 time unit (a board that forgets fast); with $W = 240$ about 10 (a board that forgets slowly). The window also changes what the controller reads, so with $W = 240$ its vote gate sees older votes.

Table 2: The four cases compared in this report. All: $q_c = 12$, $\rho \in \{1, 0.75, 0.5, 0.25\}$, $q \in \{3, 6, 12\}$, all nine rates, gate on and off, 1,000 episodes per setting.

case	budget B	window W	a post stays readable	run
gold standard	30	48	≈ 2 time units	15–16 October (`*_fine`)
$B = 100$	100	48	≈ 2	18 October (`*_budget100`)
$W = 24$	30	24	≈ 1	18 October (`*_window24`)
$W = 240$	30	240	≈ 10	18 October (`*_window240`)

The runs. The gold standard's settings at $q_c = 12$ and $\rho \in \{1, 0.75, 0.5, 0.25\}$ (from the 15–16 October runs) and the three variants on the same grid: posts read by an agent $q \in \{3, 6, 12\}$; rate $\lambda_c \in \{0.5, 1, 2, 3, 4, 5, 6, 7, 8\}$; vote gate on and off; proof stop on; 1,000 episodes per setting (444 settings per case). Unless stated otherwise:

$q = 6$, $q_c = 12$, $\lambda_c = 1$, gate on. Where the budget cannot matter, the $B = 100$ run reproduces the gold standard exactly: all silent settings, and every setting in which the gold-standard controller never reached 30 messages (113 settings).

Summary. Each variant changes one rule of the gold standard. **The gold standard's main results hold in all three**: full memory makes the swarm immune, forgetting makes it steerable, and reading more protects when agents remember and helps the controller when they forget, and η stays flat (Figures 1, 5, 7). **A budget of 100 messages lets fast controllers keep gaining**: up to +0.13 at $\rho \le 0.5$, with each extra message worth about half as much; the 30-message cap is what made the rate look irrelevant to the end point (Figures 2, 9, 10). **The board's memory protects like the agents' memory, but less**: posts that stay readable for 10 time units instead of 2 lower the false controller's gain by 0.06–0.14 at $\rho \le 0.5$; posts readable for 1 time unit raise it by up to 0.09 (Figures 1, 2). **The truth controller is the same in every case**: the vote gate keeps it mostly silent.

1 Quantities

Votes. With $n_{a,t}$ the number of agents whose last vote is answer a at time t and N_t the number of agents who have voted,

$$x_{a,t} = n_{a,t}/N_t. \tag{1}$$

At $t = 0$ nobody has voted and we set $x_{a,0} = 0$; all agents have voted by $t \approx 5$. (The 9–14 October reports divided by 24, which counted agents who had not yet acted as voting against the target; this inflated the apparent rise of the shares and the apparent bias of the controller's sensing at early times.)

Paired susceptibility. Following `BLACKBOARD_OBSERVABLES_AND_EFFICIENCIES` (§3), for a controller policy π and an answer a,

$$\chi^{(a)}_{\pi,t} \equiv G^{(a)}_t(\pi) = \mathbb{E}\big[x^{\pi}_{a,t} - x^{\text{base}}_{a,t}\big], \tag{2}$$

the mean over episodes of the difference between the controlled run and the silent run of the same episode. We report it on the controller's own target: $G_t \equiv G^{(0)}_t(\pi^+)$ for the controller targeting A0 and $G_t \equiv G^{(2)}_t(\pi^-)$ for the one targeting A2.

Target and state. The target $Z \in \{\text{A0}, \text{A2}\}$ is drawn by a fair coin. The state X at time t is the vote of a randomly chosen agent, $X \in \{\text{A0}, \text{A1}, \text{A2}\}$. Each episode e of arm s (silent 0, target A0 +, target A2 −) contributes its vote shares $y^{(e)}_s(x) = x_{x,t}$, and

$$P_s(x) = \frac{1}{E}\sum_{e=1}^{E} y^{(e)}_s(x), \qquad E = 1000. \tag{3}$$

Readability and efficiency. With $Q = \frac{1}{2}(P_+ + P_-)$ and $\mathrm{KL}(p\|q) = \sum_x p(x)\log_2[p(x)/q(x)]$,

$$I(Z;X) = \tfrac{1}{2}\,\mathrm{KL}(P_+\|Q) + \tfrac{1}{2}\,\mathrm{KL}(P_-\|Q), \tag{4}$$

the information the state carries about the target (at most 1 bit). The controllers' total change from silence splits exactly into a directed and an undirected part,

$$C = \tfrac{1}{2}\,\mathrm{KL}(P_+\|P_0) + \tfrac{1}{2}\,\mathrm{KL}(P_-\|P_0) = I(Z;X) + \mathrm{KL}(Q\|P_0), \tag{5}$$

and the efficiency is the directed share,

$$\eta = I(Z;X)/C. \tag{6}$$

Silence sometimes never shows a value of X that a controller reaches, which would make C infinite. Before computing any KL, every arm's distribution therefore receives the same pseudo-count of $\alpha = 0.5$ episodes per value of X (K values),

$$\tilde{P}_s(x) = \frac{E\,P_s(x) + \alpha}{E + \alpha K}. \tag{7}$$

With $\alpha = 1$ instead, η changes by at most 0.004; without it, by at most 0.005 where C is finite. η is not defined when $C < 0.01$ bits (no measurable change); such values are left blank. Equation (5) holds in the data to 4×10^{-15} bits.

Cost. F_t is the number of facts the controller has posted up to t (mean per episode); the gain per fact is G_t/F_t, and the marginal susceptibility is the slope between neighbouring rates,

$$\frac{\Delta G}{\Delta \lambda_c} = \frac{G(\lambda_c^{(k+1)}) - G(\lambda_c^{(k)})}{\lambda_c^{(k+1)} - \lambda_c^{(k)}}. \tag{8}$$

Episode e uses the same random draws at every rate, so the slope is estimated from per-episode differences, $[x^{(e)}_{\text{target},t}(\lambda_c^{(k+1)}) - x^{(e)}_{\text{target},t}(\lambda_c^{(k)})]/(\lambda_c^{(k+1)} - \lambda_c^{(k)})$, whose mean is Eq. (8) and whose standard error gives the 95% intervals (typically ±0.01–0.03, about 30% narrower than for two independent means).

2 Votes over time

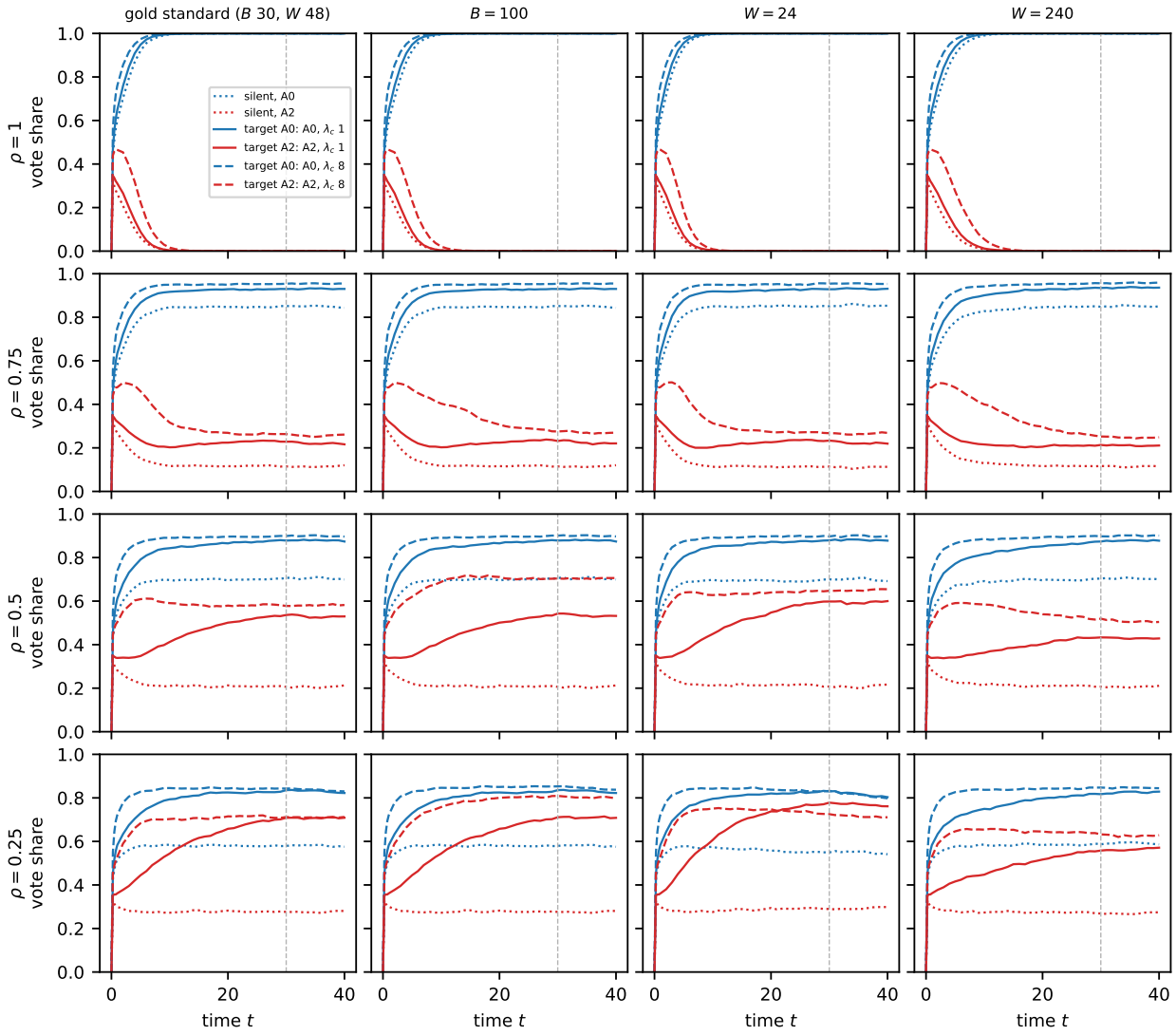


Figure 1: Share of agents voting A0 (blue) and A2 (red), Eq. (1): without a controller (dotted), with the controller targeting A0 (blue) or A2 (red) at $\lambda_c = 1$ (solid) and $\lambda_c = 8$ (dashed). Rows: memory ρ; columns: the four cases. $q = 6$, $q_c = 12$, gate on. At $t = 0$ nobody has voted and every share is 0.

- **Without a controller the board window hardly matters**: on $t = 30$ the A0 share differs by at most 0.03 between the cases (0.55–0.59 at $\rho = 0.25$, 0.70–0.71 at $\rho = 0.5$, 0.85 at $\rho = 0.75$, 1 at $\rho = 1$).
- **With full memory every case is immune**: the A2 share peaks at 0.35 ($\lambda_c = 1$) or 0.47 ($\lambda_c = 8$), the same in all four cases, and is back to 0 by $t = 30$.
- **The budget matters only for fast controllers**: at $\lambda_c = 1$ the $B = 100$ curves equal the gold standard's; at $\lambda_c = 8$ the false controller raises A2 on $t = 30$ from 0.58 to 0.70 ($\rho = 0.5$) and from 0.71 to 0.81 ($\rho = 0.25$).

- **The board window moves the false controller's result both ways**: at $\lambda_c = 1$, A2 on $t = 30$ is 0.54 (gold standard), 0.60 ($W = 24$) and 0.43 ($W = 240$) at $\rho = 0.5$; 0.71, 0.78 and 0.56 at $\rho = 0.25$.
- **The truth controller's result barely changes**: A0 on $t = 30$ stays within 0.02 of the gold standard in every case. **Every result persists** after the controller stops (from $t = 30$ to 40 the shares change by at most 0.03).

3 Observables as time series

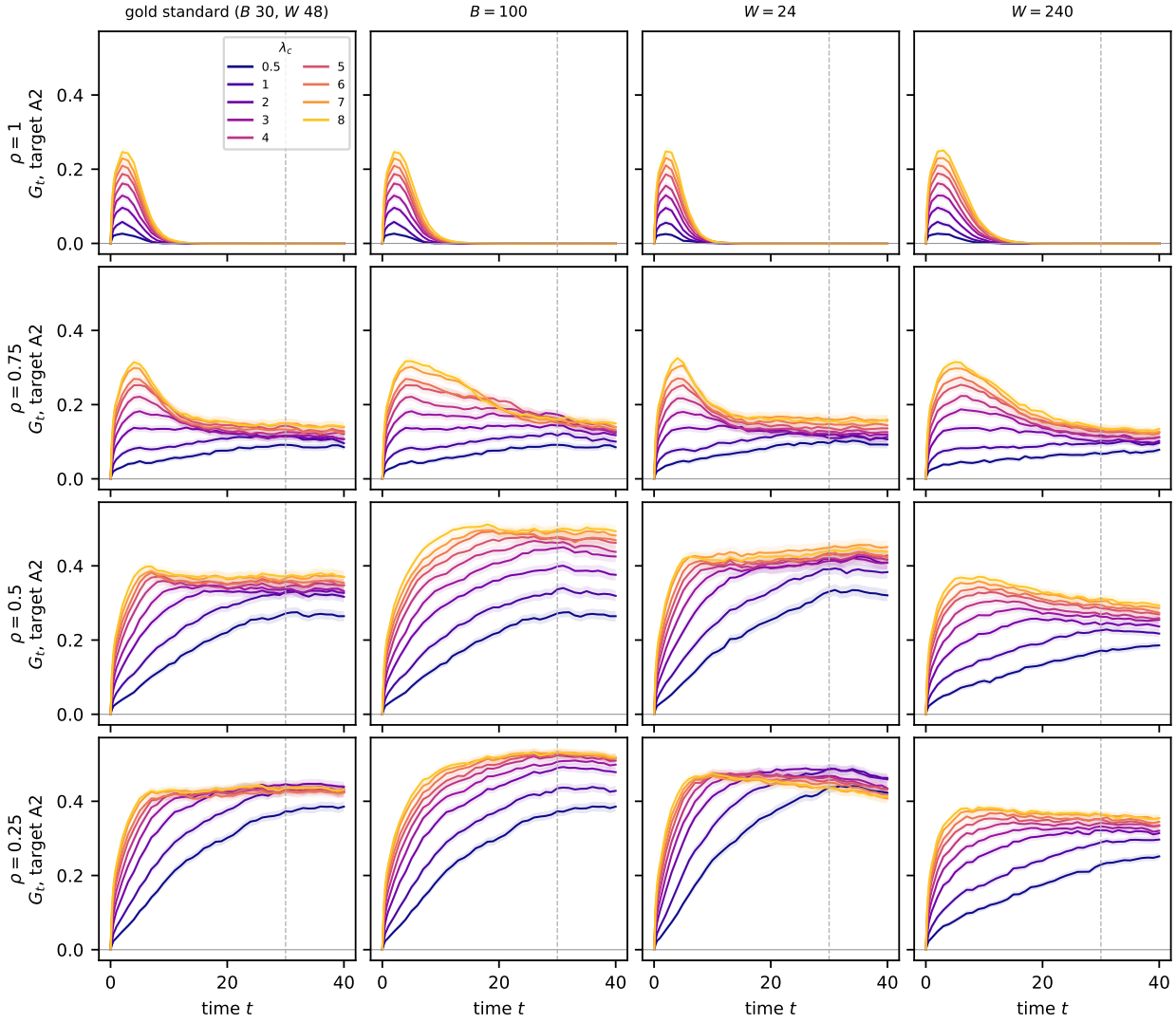


Figure 2: Paired susceptibility G_t for target A2, Eq. (2), one curve per rate λ_c (bands: 95% intervals); rows ρ, columns the four cases. $q = 6$, $q_c = 12$, gate on.

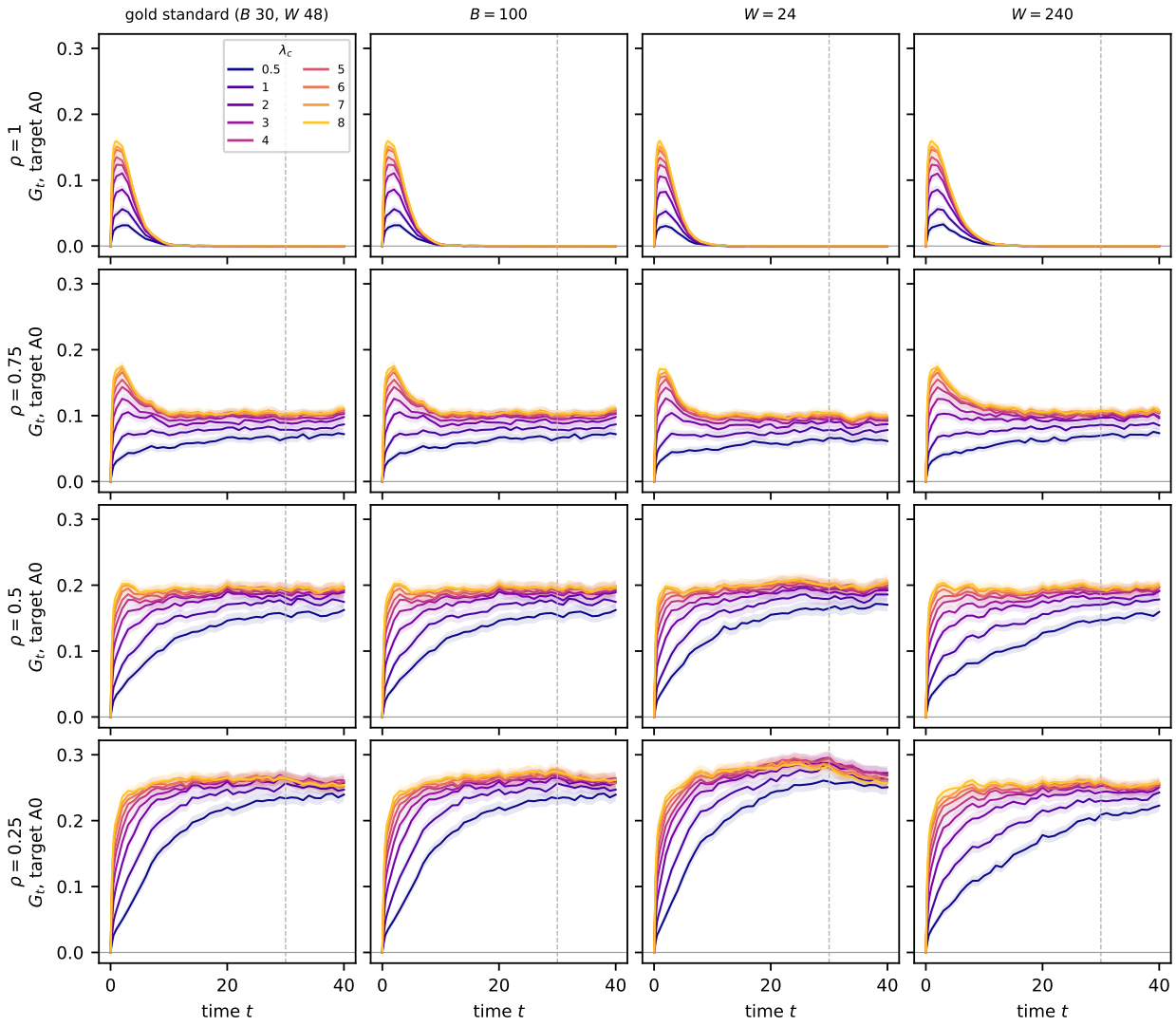


Figure 3: The same for target A0.

Susceptibility (Figures 2, 3).

- $B = 100$**: fast controllers keep gaining after the gold standard's budget would have run out.** On $t = 30$, target A2, G ranges over the rates from 0.27 to 0.50 at $\rho = 0.5$ (gold standard 0.27–0.37) and from 0.37 to 0.53 at $\rho = 0.25$ (0.37–0.45); at $\rho = 0.75$ it rises by at most 0.06, at $\rho = 1$ not at all.
- $W = 24$ **raises** G**,** $W = 240$ **lowers it**: on $t = 30$, target A2, $\rho = 0.5$: 0.33–0.45 and 0.17–0.31 against 0.27–0.37; $\rho = 0.25$: 0.44–0.49 and 0.23–0.36 against 0.37–0.45. With $W = 240$ fast controllers peak early and fade (at $\rho = 0.5$, $\lambda_c = 8$: 0.37 at $t = 5$, 0.31 at $t = 30$).
- **The truth controller is unaffected** (target A0, within 0.03 of the gold standard): the vote gate keeps it mostly silent (1–29 messages by $t = 30$), so neither a larger budget nor the window changes much.

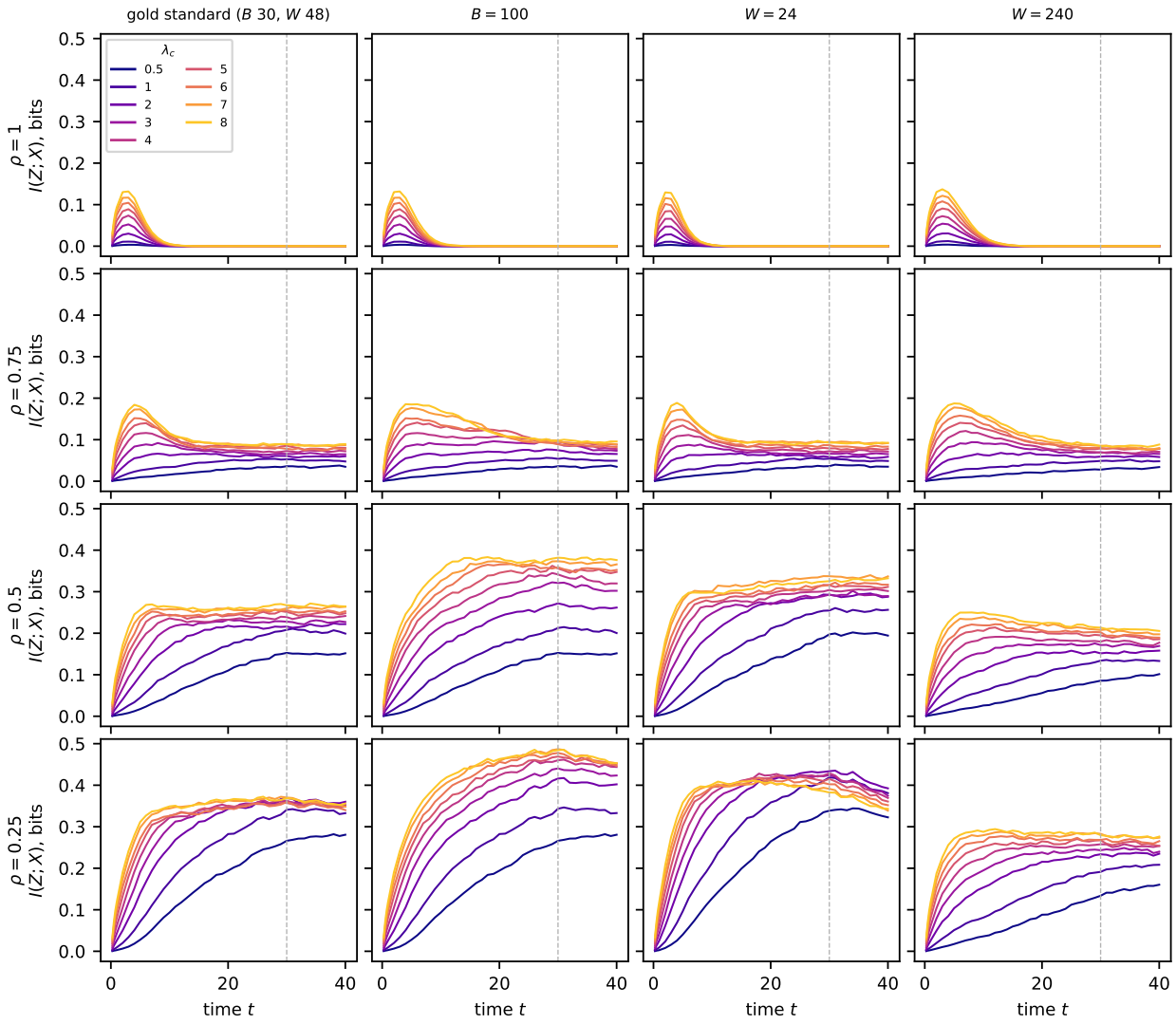


Figure 4: Readability $I(Z;X)$, Eq. (4), one curve per rate; rows ρ, columns the four cases.

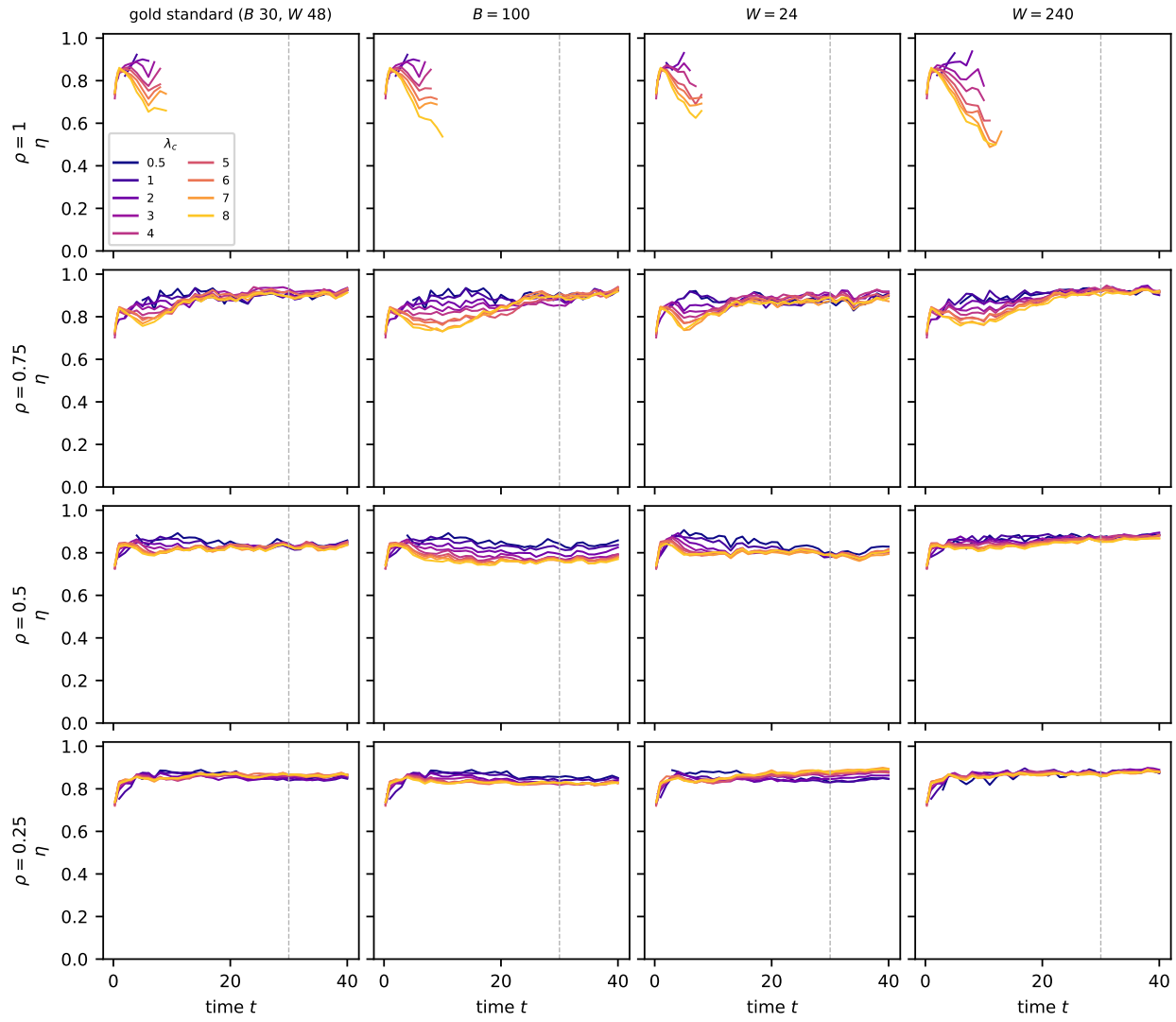


Figure 5: Efficiency η, Eq. (6), one curve per rate (blank: undefined, $C < 0.01$ bits).

Readability and efficiency (Figures 4, 5).

- **Readability follows the susceptibility in every case**: on $t = 30$, $\rho = 0.5$, $I = 0.15$–0.27 bits (gold standard), 0.15–0.38 ($B = 100$), 0.20–0.34 ($W = 24$), 0.09–0.21 ($W = 240$).
- **The efficiency does not move**: $\eta = 0.76$–0.93 on $t = 30$ in every case with $\rho < 1$ (undefined at $\rho = 1$, where neither controller changes the swarm).

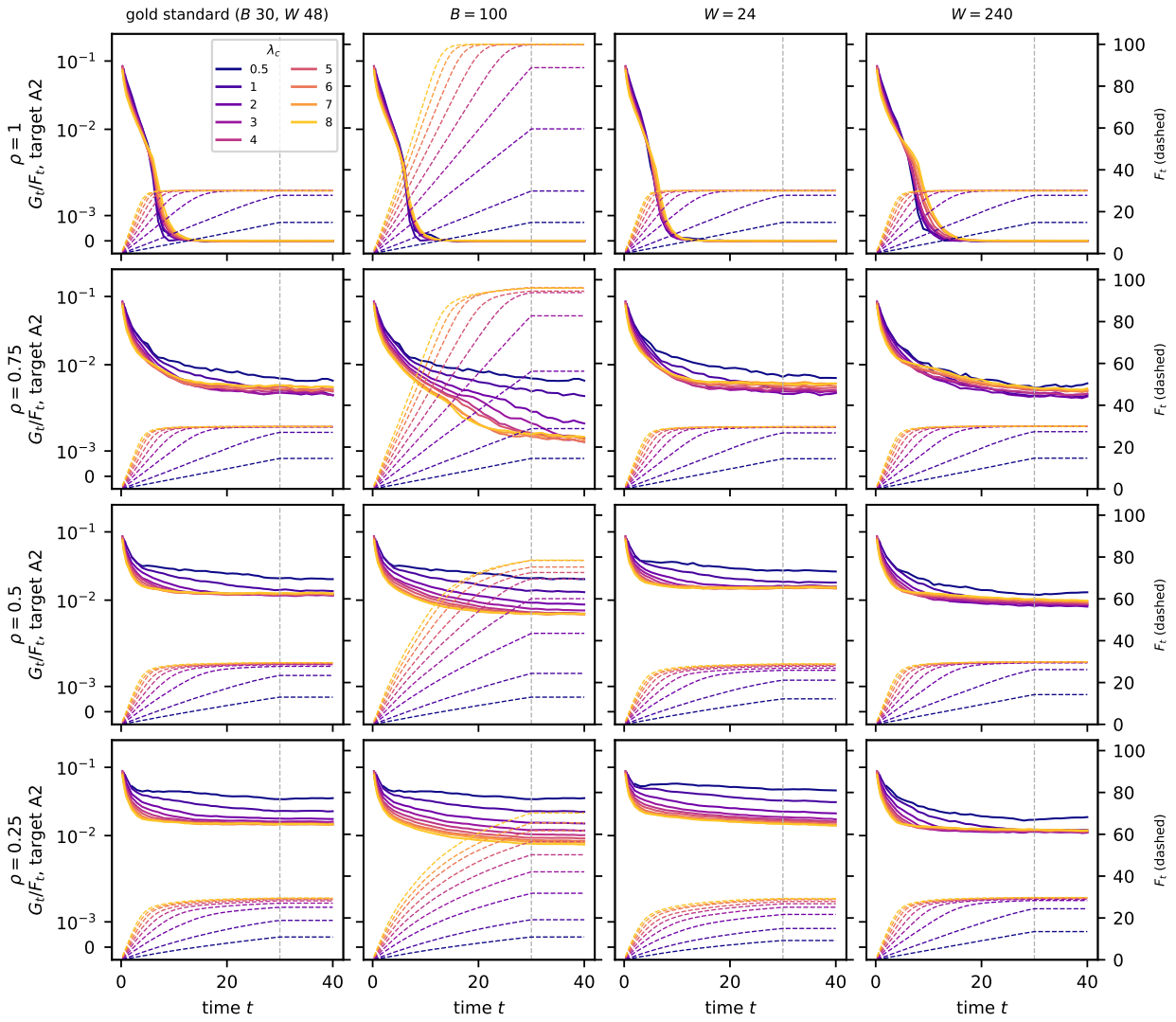


Figure 6: Gain per message G_t/F_t (solid, left axis, log scale; shown once half a message has been posted) and messages posted F_t (dashed, right axis), target A2, one colour per rate; rows ρ, columns the four cases.

Gain per message (Figure 6).

- **The extra messages of $B = 100$ are worth half as much each**: on $t = 30$, $\lambda_c = 8$, 0.0063 per message against 0.0126 at $\rho = 0.5$ (78 messages against 29) and 0.0076 against 0.0148 at $\rho = 0.25$.
- **The window changes the value of each message**: at $\lambda_c = 1$, $\rho = 0.25$: 0.033 ($W = 24$), 0.023 (gold standard), 0.012 ($W = 240$).

4 Phase diagrams

On $t = 30$, gate on; susceptibility for target A2 (grey number: its 95% interval includes 0); readability and efficiency for the pair of targets. The gold standard's third diagram ($q_c \times \lambda_c$) has no counterpart: the variants were run at $q_c = 12$ only.

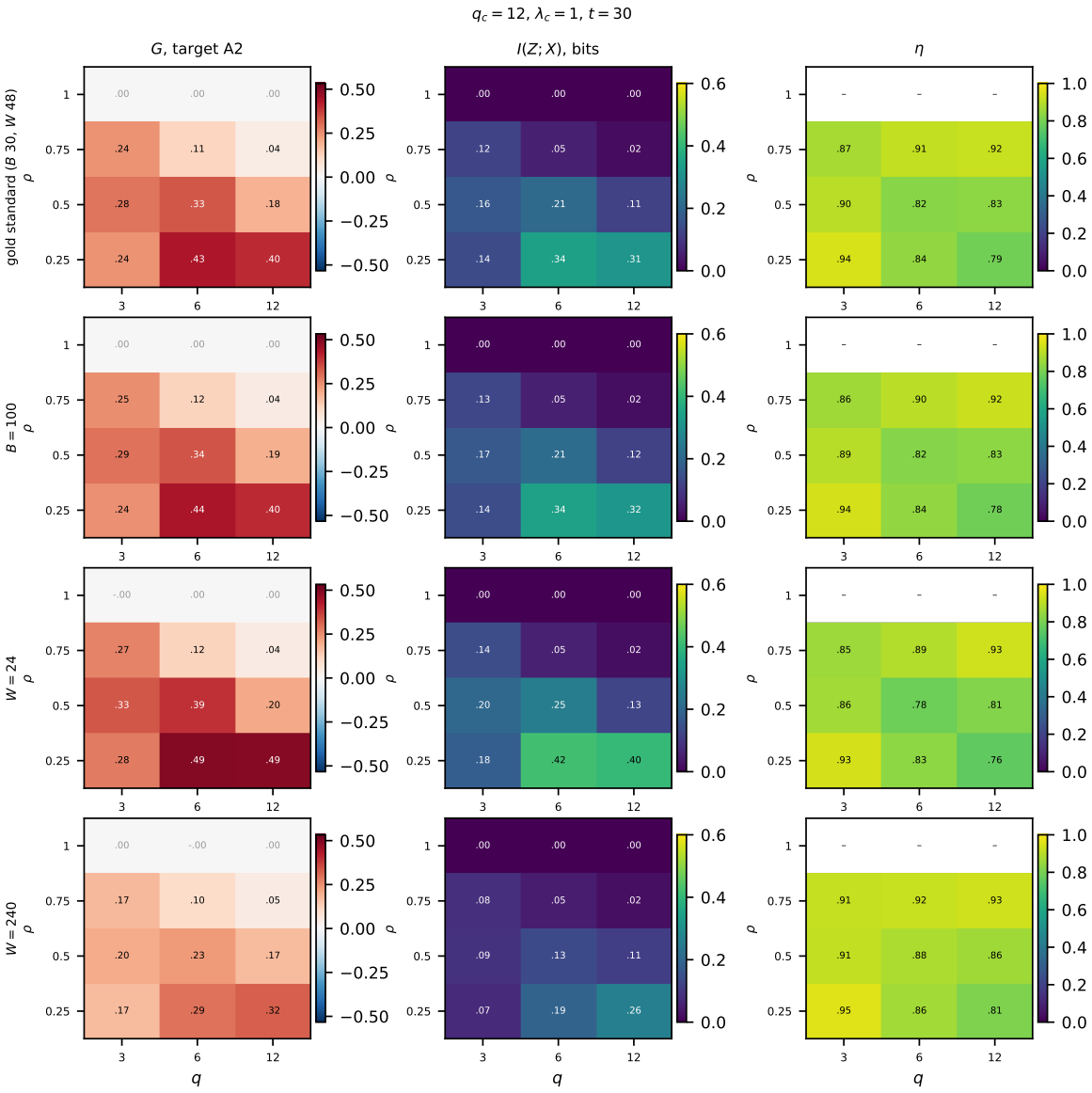


Figure 7: Susceptibility, readability and efficiency over memory ρ and the posts each agent reads q; rows: the four cases. $q_c = 12$, $\lambda_c = 1$.

- **The reading reversal survives in every case**: reading more protects the swarm at $\rho = 0.75$ (G from $q = 3$ to 12: 0.24 → 0.04 gold standard, 0.25 → 0.04 $B = 100$, 0.27 → 0.04 $W = 24$, 0.17 → 0.05 $W = 240$) and helps the controller at $\rho = 0.25$ (0.24 → 0.40, 0.24 → 0.40, 0.28 → 0.49, 0.17 → 0.32).
- $W = 240$ **makes the reversal smaller**, $W = 24$ larger; $B = 100$ leaves it unchanged at $\lambda_c = 1$.

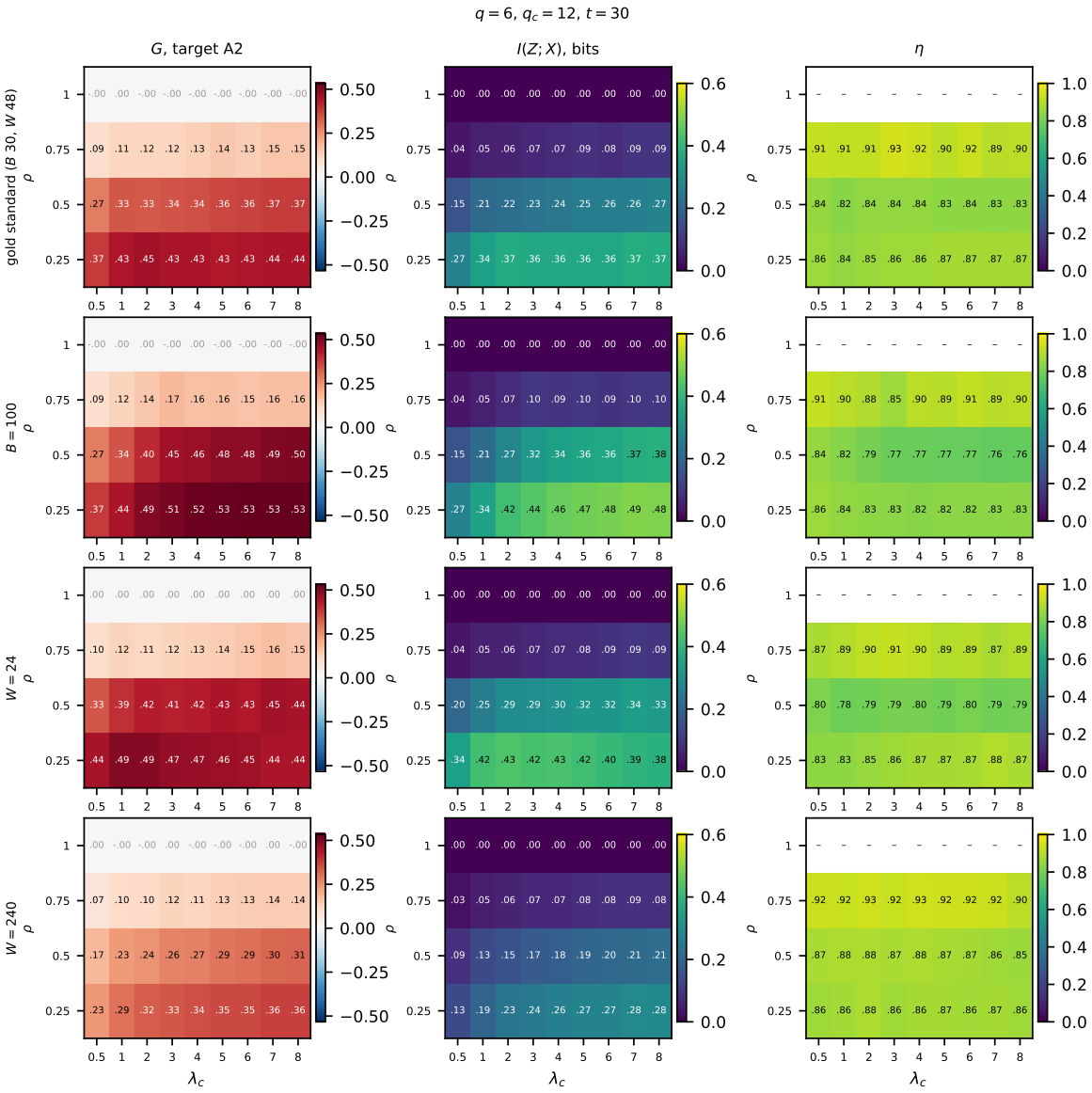


Figure 8: Susceptibility, readability and efficiency over memory ρ and the rate λ_c; rows: the four cases. $q = 6$, $q_c = 12$.

- **Memory stays the main axis in every case**: each row is near zero at $\rho = 1$ and largest at $\rho \leq 0.5$. $B = 100$ **adds the strongest rate gradient**: from $\lambda_c = 1$ to 8, G grows by 0.16 ($\rho = 0.5$) and 0.10 ($\rho = 0.25$), against 0.04 and 0 in the gold standard; $W = 240$ adds a weaker one (0.09, 0.07), $W = 24$ none (0.05, −0.05).

5 Susceptibility against the rate, against messages, and its slope

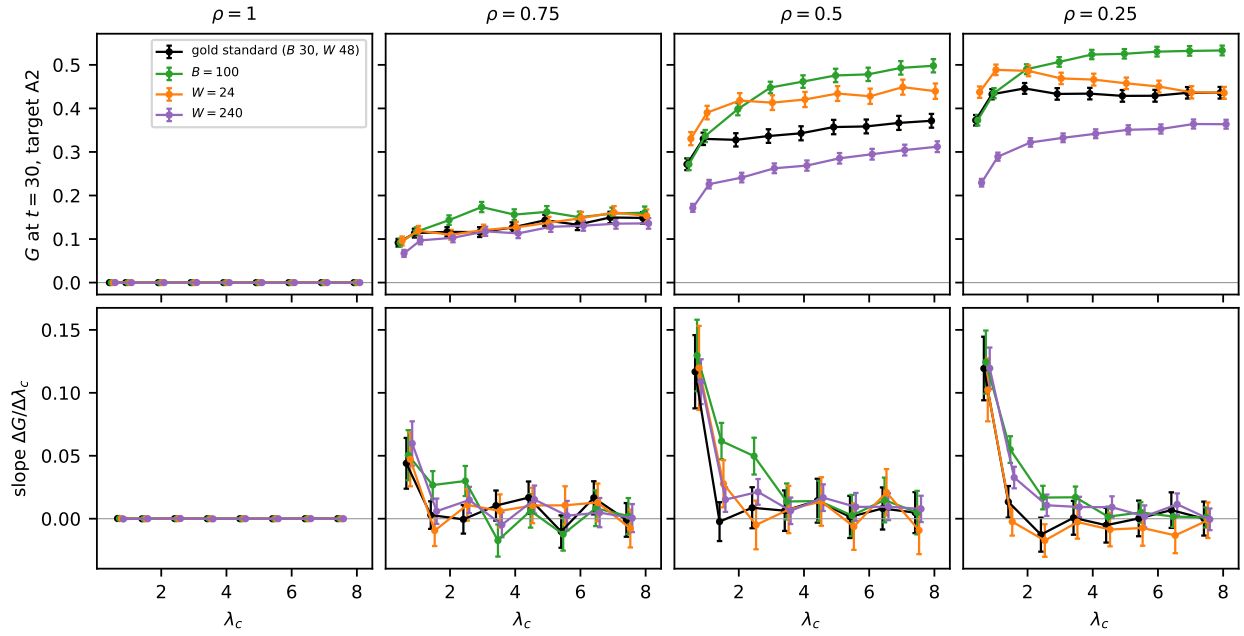


Figure 9: Top: G on $t = 30$, target A2, against the rate λ_c, one colour per case (bars: 95% intervals; points shifted slightly sideways). Bottom: the slope between neighbouring rates, Eq. (8), with paired 95% intervals. $q = 6$, $q_c = 12$, gate on.

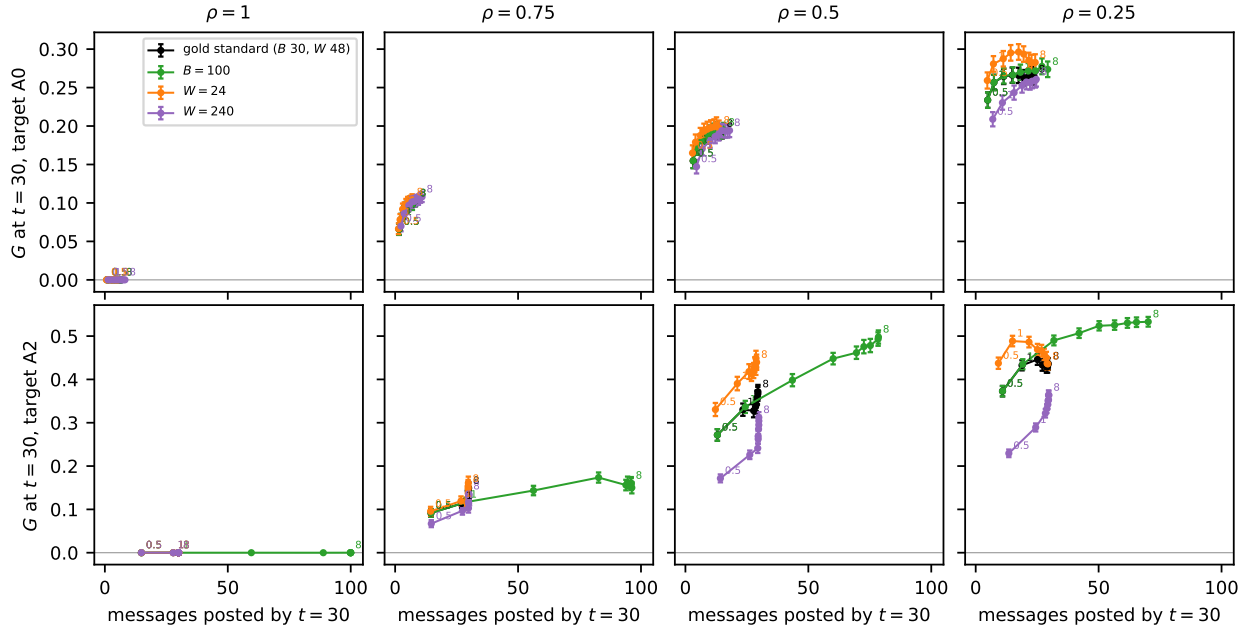


Figure 10: G on $t = 30$ against the messages posted by then, one point per rate (labels: $\lambda_c = 0.5$, 1, 8), target A0 (top) and A2 (bottom), one colour per case; bars: 95% intervals. Messages can repeat a fact: the pool holds 12.

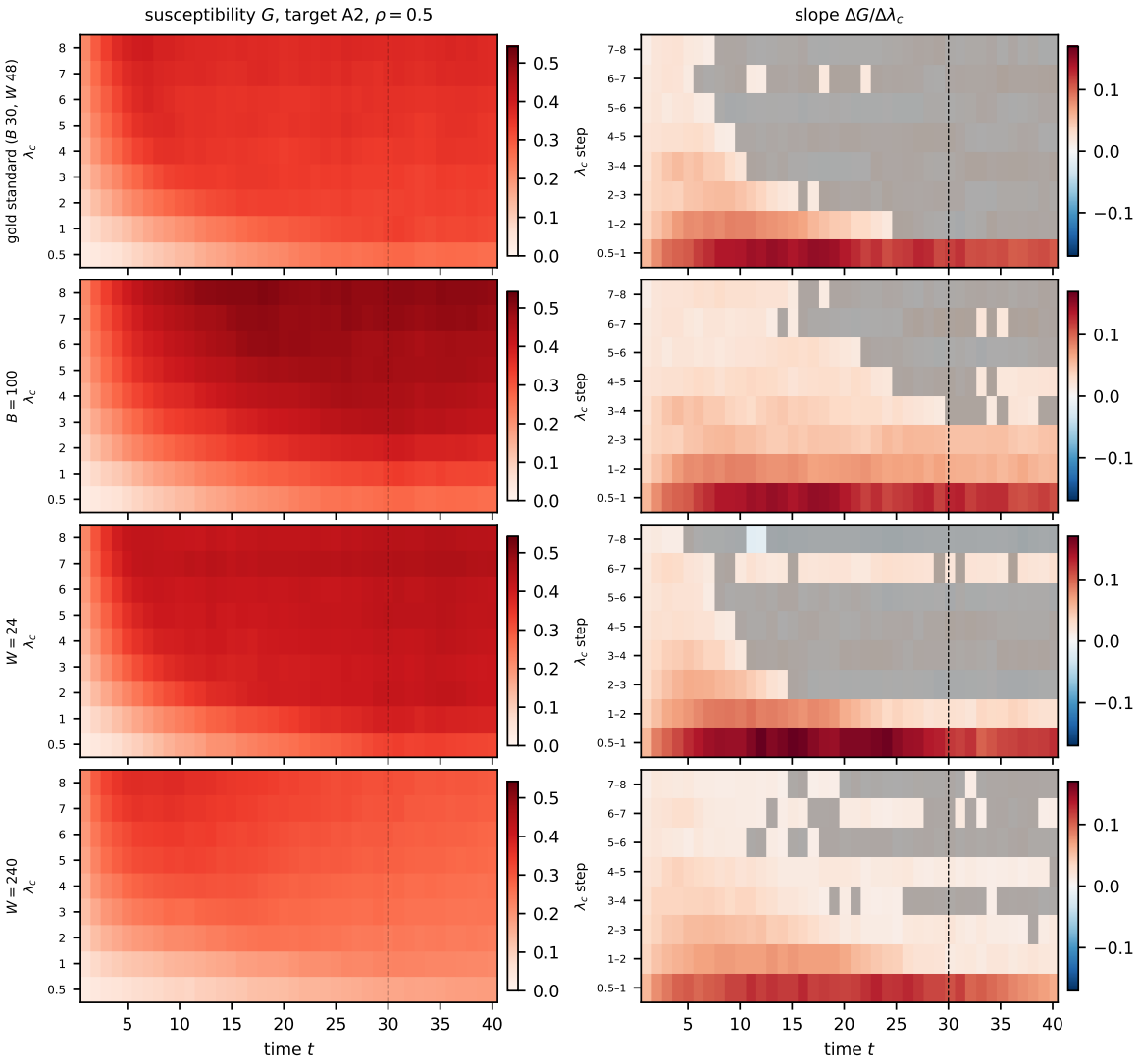


Figure 11: G, target A2 (left), and its slope between neighbouring rates (right) over time (x) and rate (y), $\rho = 0.5$; rows: the four cases. Faded cells: the paired 95% interval of the slope includes 0.

- **With $B = 30$ only the first step of rate pays on** $t = 30$ (gold standard, $\rho = 0.5$: +0.12 per unit of rate from 0.5 to 1); **with $B = 100$ the next steps pay too**: +0.13, +0.06, +0.05 up to $\lambda_c = 3$ at $\rho = 0.5$, and four significant steps at $\rho = 0.25$ (Figure 9).
- **The budget moves the controller along one curve; the window moves the curve** (Figure 10): with $B = 100$ the false controller goes from 0.33 (24 messages) to 0.50 (78) at $\rho = 0.5$; at the same 28–30 messages G is 0.45 ($W = 24$), 0.37 (gold standard) and 0.31 ($W = 240$).
- **Over time** (Figure 11): in the gold standard the slopes above $\lambda_c = 1$ fade between $t \approx 5$ and 25, and on $t = 30$ only the first step is significant; with $B = 100$ the steps up to $\lambda_c = 3$ stay significantly positive to $t = 30$.
- **The gate matters little in every case**: gate off minus on, target A2, $t = 30$, is between −0.03 and +0.04.

6 Who holds the proof

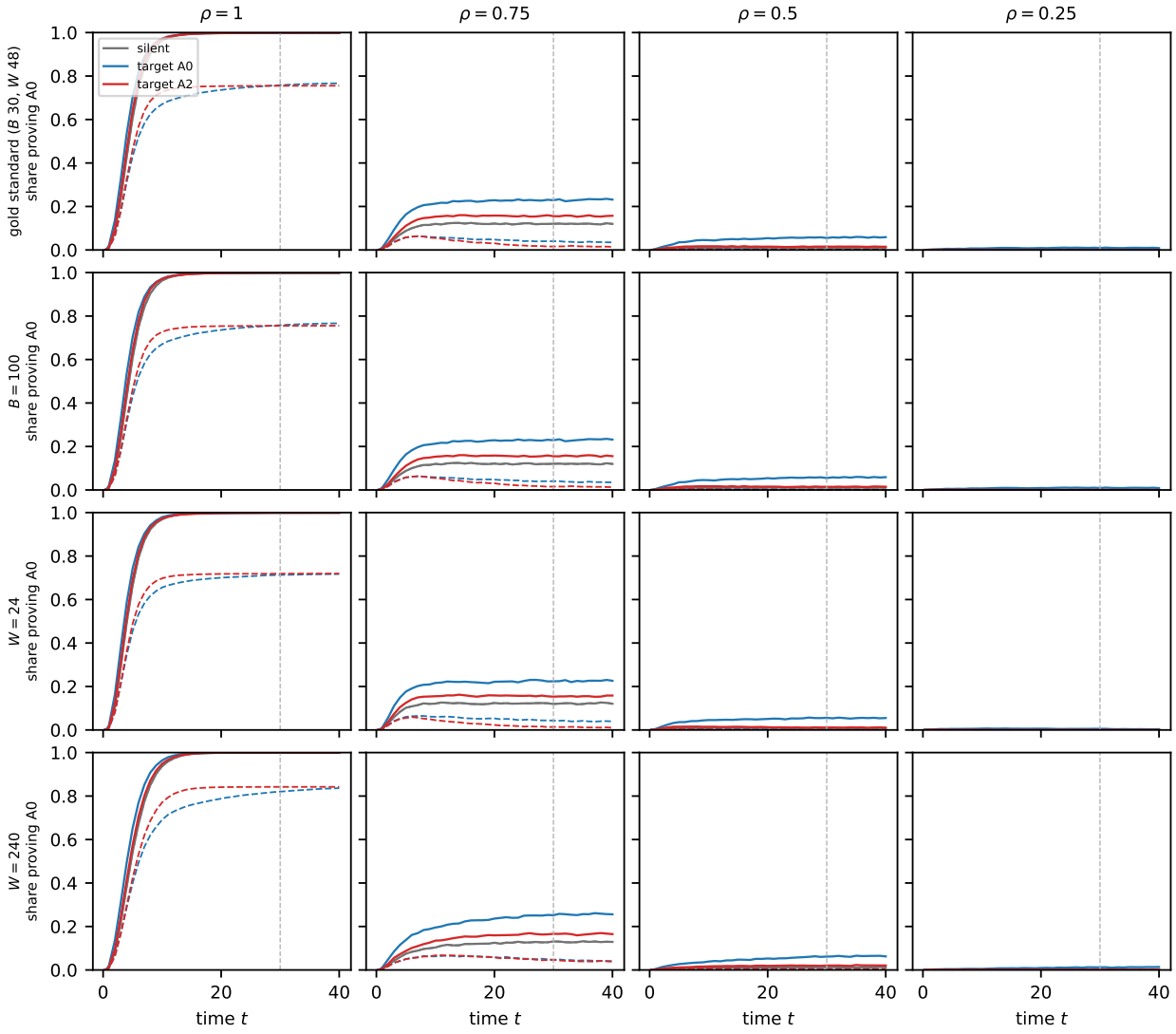


Figure 12: Share of agents whose remembered facts prove A0 (solid) and who prove it from their original facts only (dashed): silent (grey), target A0 (blue), target A2 (red), $\lambda_c = 1$; rows: the four cases, columns ρ.

- **Only full memory keeps the proof**, in every case: at $\rho = 1$ all agents prove A0 on $t = 30$; at $\rho = 0.75$ 12–25%; at $\rho \leq 0.5$ at most 6%.
- **The window changes how many proofs stay the agents' own** at $\rho = 1$: with a controller, 0.71–0.72 ($W = 24$), 0.76 (gold standard and $B = 100$), 0.82–0.84 ($W = 240$), against 1 without one.

7 Conclusions

1. **The main results of the gold standard hold in all three variants**: full memory makes the swarm immune; forgetting makes it steerable; reading more protects when agents remember and helps the controller when they forget; η stays at 0.76–0.93; the gate matters little.
2. **A larger budget raises the end point, with diminishing returns.** $B = 100$ adds up to 0.13 to the false controller's gain on $t = 30$ where the swarm is already steerable ($\rho \leq 0.5$), up to 0.06 at $\rho = 0.75$ and nothing at $\rho = 1$; each extra message is worth about half as much. The gold standard's "the budget sets the end point and the rate sets the speed" becomes: *higher rates and budgets both raise the end point, with diminishing returns per message; the 30-message cap is what made the rate look irrelevant.*
3. **The board's memory protects like the agents' memory, but less.** Keeping posts readable for 10 time units instead of 2 lowers the false controller's gain by 0.06–0.14 at $\rho \leq 0.5$ (every rate); keeping them

1 time unit raises it by up to 0.09. Memory ρ moves it by up to 0.43. A likely reason, not tested: a long window keeps the agents' older posts, which carry the facts of the proof, readable for longer.

4. **The truth controller is the same in every case**: the vote gate keeps it mostly silent.

Caveats. The window changes what the controller reads too: with $W = 240$ its vote gate sees votes up to 10 time units old, and at $\lambda_c = 1$ it posts 24–28 messages instead of 19–28. The variants were run at $q_c = 12$ and four memory values only. One world, 24 exact-Bayesian agents, one controller policy; 1,000 episodes per setting (95% intervals on G about ± 0.02). Data: `results/simulation_2/2026-10-18_sim2_symmetric_*`; tables: `results/gold_standard/2026-10-18/` (variants) and `2026-10-16/` (gold standard); code: `shared_analysis/gold_standar`